Analyzing Vaccine Discourse in YouTube Comment Sections- A Study of COVID-19 Vaccination in India

Abstract

The COVID-19 pandemic brought unprecedented challenges globally, particularly in the development and acceptance of vaccines in a short period. Social media platforms like YouTube became critical spaces where vaccine-related discussions unfolded, influencing public opinions and emotions. In the Indian context, where large-scale vaccination campaigns were rolled out, achieving high vaccination coverage by early 2023. But still, there is a discussion going on about the COVID-19 vaccines, and whether they are safe and effective for the vast majority of people. In these scenarios understanding public discourse is vital. This study analyzed comments on YouTube videos related to COVID-19 vaccination to explore the sentiments, emotions, and stances expressed by users.

This study analyzed comments on YouTube videos related to COVID-19 vaccination, employing a combination of manual observation, data extraction using the YouTube API, and advanced large language model (LLM) processing techniques. The study categorized 4856 comments including replies into positive, negative, and neutral sentiments, and also analyzed the emotional aspects of the text, identifying anger as the most dominant emotion. Anti-vaccine narratives were strongly associated with emotions like anger and fear, often fueled by misinformation. Additionally, the percentage of individuals holding anti-vaccine opinions is almost double the number of those with pro-vaccine opinions. The findings highlight the importance of addressing vaccine hesitancy, countering misinformation, and fostering trust through transparent communication. This pilot study provides actionable insights for improving vaccine acceptance, particularly in India, by emphasizing the need for targeted public health campaigns and the role of social media discourse in shaping public opinion.

Introduction

Vaccines are one of the significant achievements in modern medicine, playing a critical role in preventing infectious diseases and saving millions of lives worldwide. They have become a cornerstone of public

health, significantly reducing the burden of infectious diseases by stimulating the body's immune response. While vaccine hesitancy is perceived to be a long-standing problem, it has recently become a great urgency in the COVID-19 pandemic. "In 2019, the WHO listed vaccine hesitancy as one of the top 10 threats to world health" (Cascini, 2022). Vaccination programs have been central to public health strategies globally. Particularly in India, vaccines have played a crucial role. The Universal Immunization Programme (UIP) has been instrumental in providing vaccines to millions, achieving in eradicating and preventing diseases like smallpox, polio (poliomyelitis) and other vaccine preventable diseases (VPDs). (Lahariya, 2014). Recently, to fight against the coronavirus, India's swift rollout of COVID-19 vaccines, including Covishield (AstraZeneca) and India's homegrown Covaxin, was a landmark effort. By "6 January 2023, India has been able to administer more than 220 crore COVID-19 vaccine doses across the country" (MoF, 2023).

Despite the success of vaccination programs, public discourse around vaccines remains polarized, especially on social media. Social media platforms have become central to the dissemination of information and opinions on vaccines. YouTube serves as a major platform for information sharing worldwide, it allows individuals and organizations to share their perspectives on critical issues including the benefits, risks, and challenges of COVID-19 vaccination. However, the platform's comment sections have also become fertile ground for debates, misinformation, and polarized discussions, particularly during the COVID-19 pandemic (Gupta, 2022). In the Indian context, vaccine discourse reflects the country's unique socio-cultural, political, and economic dimensions. Supporters of vaccines emphasize the achievements of India's immunization programs, the role of indigenous research in developing COVID-19 vaccines, and the importance of achieving immunity to curb pandemics. On the other hand, vaccine scepticism often stems from mistrust in government and pharmaceutical companies, misinformation about vaccine safety and side effects, and rumours regarding efficacy. This scepticism was particularly evident during the early phases of the COVID-19 vaccination drive, where misinformation about the safety of Covishield and Covaxin circulated widely on social media (Gupta, 2022).

This research project aims to analyse the discourse around COVID-19 vaccination in YouTube comment sections within the Indian context. By examining comments on 6 videos, we employ aspect sentiment analysis by using a Large Language Model to identify recurring aspects - keywords/ themes, the prevalence of misinformation, and the dominant sentiments shaping public opinion and emotions. The analysis focuses on how YouTube serves as a platform for both positive engagement and the spread of vaccine hesitancy. The findings of this study will provide insights into how digital platforms influence

vaccine acceptance in India, offering valuable lessons for public health communication. In our pilot study, we are analysing a small sample size of videos. After establishing the protocol for analysis, we aim to comprehensively work on this research and conduct an in-depth analysis of the YouTube comments section discourse on COVID-19 vaccination.

Hypothesis

This pilot study aims to examine the dynamics of vaccine discourse on YouTube in India through a qualitative and quantitative analysis of user comments. Specifically, we hypothesize the following-

1. **Aspects of Discussion**- YouTube comments on vaccine-related videos predominantly focus on key topics such as safety concerns, vaccine efficacy, political influence and societal implications.
2. **Emotional and Sentiment Patterns**- The discourse reflects a diverse range of emotions and sentiments, with distinct patterns in pro-vaccine and anti-vaccine comments. Pro-vaccine comments are expected to exhibit positive emotions like hope and happiness, while anti-vaccine comments are likely associated with fear, anger, and mistrust.
3. **Polarization of Opinions**- Vaccine-related discussions on YouTube are polarized, with clear divisions between supportive and opposing views. Anti-vaccine narratives tend to be more emotionally charged and driven by concerns over misinformation and distrust, whereas pro-vaccine narratives emphasize benefits and collective well-being.

By exploring these aspects, the study aims to understand how YouTube shapes public opinion on vaccination and contributes to the polarization of vaccine-related discourse. These insights could inform strategies to address vaccine hesitancy and enhance public health communication.

Methodology

This study employs a mixed method approach, using both qualitative and quantitative techniques to analyze YouTube comments related to COVID-19 vaccination. We combined manual observations, data extraction using the YouTube API, and advanced text processing through large language models to understand vaccine discourse.

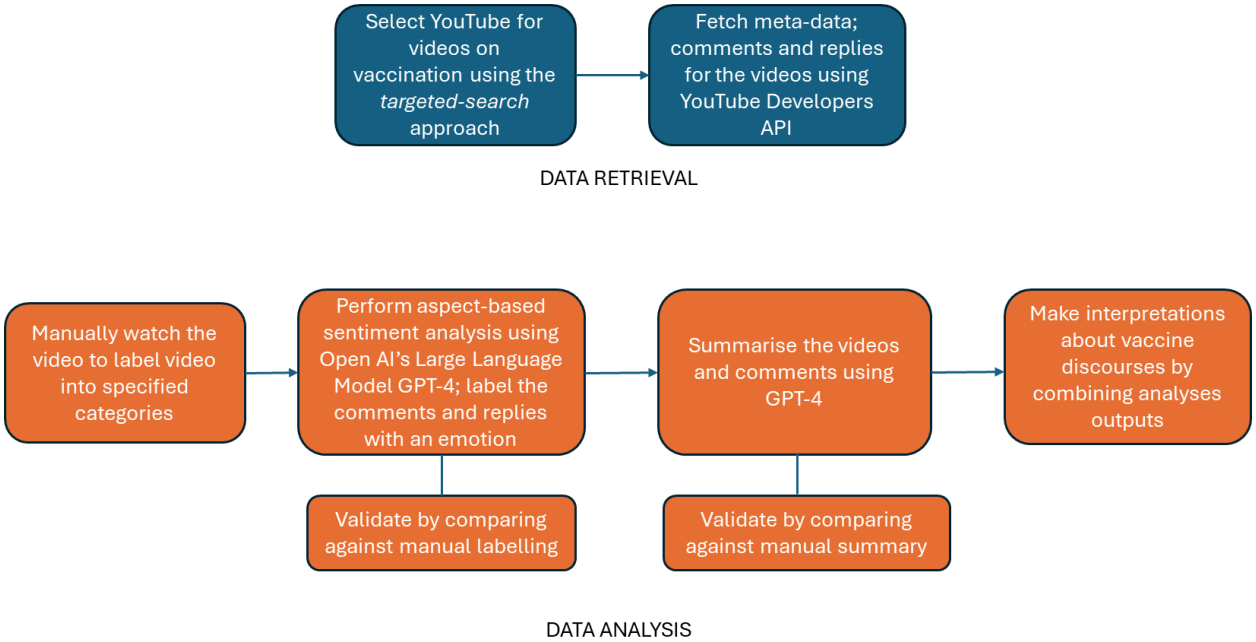


Fig. Methodology Flowchart

The methodology is outlined in the following sections-

Video Selection Process

To ensure a representative sample of videos discussing COVID-19 vaccines and general discussion on vaccines, we employed a search-based approach. The search terms were carefully curated based on our preliminary observations of common phrases and topics related to vaccine discourse as well as played with different keywords and phrases to get various types of videos on the discourse. Keywords/ phrases included-

General terms- "vaccine", "covaxin", and "covishield"

Public concerns- "rare side effects of vaccine"

Misinformation triggers- "rare side effects of covaxin", and "rare side effects of covishield"

For each keyword/ phrase, we retrieved the top two videos based on YouTube API's search. We specifically targeted videos on the above terms, covering the Indian context of COVID-19 vaccination.

Data Retrieval

Manual Process - The data collection process was twofold. First, each selected video was manually watched to understand its content and tone, This allowed us to contextualize the comments, better interpret the sentiment behind them and label the video into a specific category. As well as collect all metadata including title, channel, description, time/ length (in min), views, likes, published date, language-spoken, no-of-comments, video-category, video-type. wrote a summary as well as generated the summary using AI.

YouTube API - To extract comments and comment threads and associated metadata, we used the YouTube developer API v3. We identified 09 videos matching our keywords using the API's search functionality. For each video, we extracted the top 100 highly engaged comments based on the number of likes and their replies.

List of the Videos and Metadata

title	video-retrieval	channel	time/length (in min)	video-category	views	likes	published-date	no-of-comments	language-spoken
Are Coronavirus Vaccines Safe? | Blood Clotting Side Effects | Dhruv Rathee	"rare side effects of covaxin"	Dhruv Rathee	8:37	Youtube Influencer	4329371	211000	4/12/2021	17999	english
Covishield Side Effects :कोविशील्ड Vaccine से हो सकता है हार्ट अटैक! | Heart Attack | AstraZeneca	"covishield"	GPAT DISCUSSION CENTER	1:20	Educational	108967	1200	5/2/2024	159	Hindi
Covishield Vaccine Side Effects: AstraZeneca Admits Its Covid Vaccine Can	"rare side effects of covishield"	India Today	8:02	network news	88777	530	5/1/2024	207	english

Cause Rare Side Effects									
India: Lapses in BHU research on Covid-19 vaccine Covaxin: Bharat Biotech | WION	"covaxin"	WION - The World is One News	1:56	network news	1615	18	5/17/2024	8	english
New study finds rare side effects from COVID-19 vaccine	"rare side effects of vaccine"	Global News	0:33	network news	365380	5100	02/24/2024	2245	english
Study reveals AstraZeneca and Pfizer vaccines linked to rare neurological side effects	"rare side effects of vaccine"	Arirang News	0:40	network news	1500000	15000	10/27/2021	6169	english
Study: Covaxin developed by Indian Pharma firm Bharat Biotech neutralizes UK variant | COVID-19	"covaxin"	WION - The World is One News	1:42	network news	212759	7800	1/27/2021	870	english
Watch: PM Modi gets Bharat Biotech's Covaxin as India widens vaccination drive	"covaxin"	Hindustan Times	1:31	network news	102113	2100	Mar 1, 2021	598	English

Aspect-based Sentiment Analysis and LLM Processing-

Once the comments were collected, we employed the OpenAI API (LLM) for advanced aspect-based sentiment analysis of comments text. The analysis was guided by a thorough prompt, which helped in

categorizing each comment effectively. Due to the constraint of API use and time for the pilot, we analysed 4856 comment text. We engineered the prompt to do the following:

1. **Aspect Identification-** Each comment was analyzed to determine whether it was directly related to vaccination, COVID-19, or infectious diseases. Comments irrelevant to these topics were categorized as "unclear." This step ensured the analysis focused only on discourse pertinent to the research objective.
2. **Sentiment Classification**- Comments were categorized into one of three sentiment categories-
    - **Positive-** Reflecting supportive, optimistic, or approving attitudes.
    - **Negative-** Indicating concerns, disapproval, or criticisms.
    - **Neutral-** Expressing neither strong positive nor negative sentiment, often factual or informational in tone.
3. **Emotion Tagging -** Each comment was tagged with one or more of the following emotions based on its tone and content- happiness, sadness, fear, anger, surprise, or disgust. If it is unclear, it was labelled as unclear.
4. **Vaccination View-** Comments were classified based on their stance toward vaccination into three categories-
    - **Pro-Vaccine-** Advocating for or expressing support for vaccination.
    - **Anti-Vaccine-** Expressing hesitancy, opposition, or distrust in vaccination.
    - **Unclear-** When the stance was ambiguous or not directly addressed.

Example Prompt- the example of the prompt shows how it works and gives us an analysis.

Comment Text- “Swelling in the brain is a lot worse than 2 weeks of a 'bad cold'”

Aspect- vaccine side effects

Sentiment- Negative

Emotion- Fear

VaccinationView- anti-vaccine

We tested transformer-based sentiment analysis tools. However, we found them limiting for our purposed

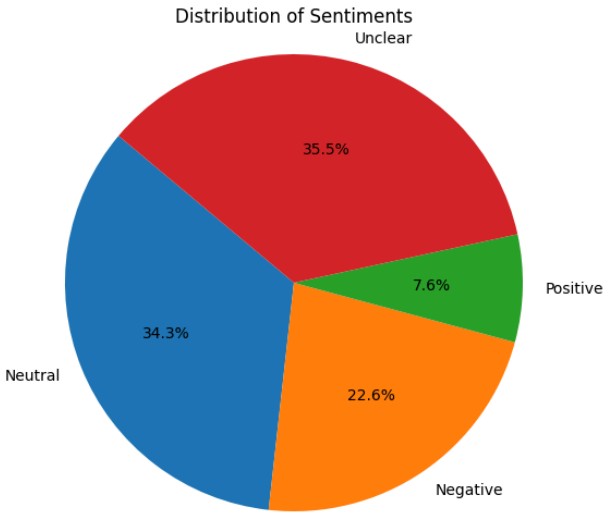
Visualization Techniques

To present findings on COVID-19 vaccine discourse from YouTube comments. We used Pie charts that displayed aspect-based sentiment analysis— sentiment distribution, emotion distribution and vaccination views distribution.

Results

The analysis of the 4856 comments (15-30 comments and comment threads) from 8 YouTube videos related to COVID-19 vaccination, using aspect-based sentiment analysis and emotional tagging, revealed several key findings about public discourse. The data highlights a mix of sentiments, emotions, and vaccination views, with polarized perspectives on the subject.

Aspect-Based Sentiment Analysis-



The sentiment analysis revealed the following distribution of sentiments:

- **Neutral Sentiments:** A total of 1,668 comments (35.5%) expressed neutral sentiments. These comments often indicated a lack of strong opinions on vaccination.
- **Negative Sentiments:** A total of 1,096 comments (22.6%) exhibited negative sentiments, associated with concerns about vaccine safety, misinformation about side effects, and mistrust in government actions.
- **Positive Sentiments:** 369 comments (7.6%) were classified as positive, generally supporting vaccines, emphasizing their importance for public health and personal safety.

  A large portion of the comments, 1,723 (34.3%), were classified as unclear, reflecting ambiguity or confusion about the comment text.

The distribution of sentiments underscores the polarized nature of vaccine discourse, with a predominance of negative and neutral comments. The prevalence of unclear comments could also indicate the presence of misinformation or a lack of clear, reliable information about vaccines.

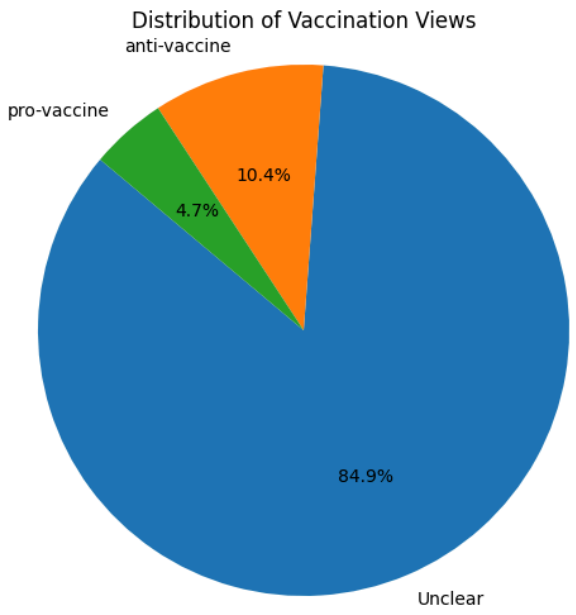
Emotional Tagging

The emotional content of the comments varied widely, with some dominant emotions reflecting concerns over vaccine safety and others indicating hope or frustration. The most frequent emotional responses included:

- **Anger:** 663 comments (13.6%) were tagged with anger, often reflecting concerns about vaccine mandates, side effects, or mistrust in the vaccination process.
- **Happiness:** 456 comments (9.3%) expressed happiness, typically tied to the relief of vaccination availability or personal experiences of positive outcomes.
- **Fear:** 123 comments (2.5%) expressed fear, primarily associated with anxiety about potential side effects or the speed of vaccine development.
- **Sadness:** 232 comments (4.7%) were linked to sadness, often stemming from negative personal experiences or frustration with the vaccination process.
- **Disgust:** 118 comments (2.4%) indicated disgust, frequently in the context of misinformation or negative experiences related to vaccines.
- **Other Emotions:** Various other emotions were noted, including frustration, skepticism, pride, and concern.

The emotional landscape of these comments clearly illustrates the emotional complexity surrounding vaccine discussions. The prominence of anger and fear points to a strong emotional engagement with the topic, often tied to misinformation or negative personal experiences.

Vaccination View



The analysis of vaccination views revealed a notable polarization:

- **Anti-Vaccine Comments:** 506 comments (10.5%) expressed opposition to vaccination. These comments were often linked to fears about vaccine safety, conspiracy theories, and distrust in government and pharmaceutical companies.
- **Pro-Vaccine Comments:** 226 comments (4.6%) supported vaccination, with comments focusing on the importance of vaccines for public health and individual safety.

A total of 4,124 comments (85%) did not clearly express either pro- or anti-vaccine views, underscoring the mixed and uncertain opinions expressed by users.

Discussion

The results of this study strongly support the hypothesis that vaccine discourse on YouTube remains highly polarized. The analysis of comments related to COVID-19 vaccination, particularly in the Indian context, reveals a complex and emotionally charged public discussion that includes both support and significant opposition to vaccines.

Polarization of Vaccine Sentiment

One of the most striking findings from the sentiment analysis is the polarization of vaccine-related discourse. The overwhelming majority of comments expressed either neutral or negative sentiments, with only a small portion expressing positive sentiments. This suggests that while the majority of the population may not be strongly opposed to vaccines, a significant portion remains sceptical or concerned, often due to misinformation or emotional reactions such as fear and anger.

This finding is consistent with existing literature that suggests social media platforms like YouTube are highly influential in shaping public opinion but also contribute to the spread of misinformation and vaccine hesitancy. The high volume of negative sentiments and emotions like anger and fear aligns with findings from studies indicating that vaccine hesitancy is often driven by misinformation about vaccine side effects and safety (Gupta, 2022). Additionally, the negative emotional responses could be linked to public anxiety about government mandates and the rapid rollout of COVID-19 vaccines (Cascini, 2022).

Emotional Complexity

The emotional tagging revealed that anger (13.6%) and happiness (9.3%) were the two most dominant emotions in the comments, which contrasts sharply. While anger was often related to fears of adverse effects, conspiracy theories, and governmental pressure, happiness generally reflected relief over vaccine availability and personal experiences of vaccination. This emotional contrast suggests that the vaccine debate is not only about facts but also heavily influenced by emotional responses and personal experiences.

Fear, while less prevalent than anger, remains a critical emotion, as it signals concerns about the unknown risks of vaccines and the speed at which they were developed (Teng et al., 2021). The presence of fear and sadness also highlights the emotional toll of vaccine-related discourse, especially among those who have been exposed to negative or alarming content online.

Prevalence of Anti-Vaccine Sentiment

The prominence of anti-vaccine comments (10.5%) in the comment sections of YouTube videos is significant. While not directly searching for anti-vaccine content, the analysis found that **anti-vaccine sentiments were more than twice as frequent as pro-vaccine sentimentaas** (4.6%). This highlights the challenges faced by public health authorities in countering misinformation and the spread of vaccine-related conspiracy theories, especially on platforms like YouTube.

Future Directions

While this pilot study provides valuable insights into the emotional and sentiment-driven nature of vaccine discourse on YouTube, several directions remain for further research:

1. **Getting a large number of videos and all the Comments and comment threads-** Future research should expand the analysis to include all comments from selected videos, providing a more comprehensive dataset. This will allow for more robust statistical analysis and reduce potential bias from a limited sample size.
2. **Manual Analysis and comparison-** Incorporating manual coding of comments will help capture nuanced emotions and contextual meanings that automated models may miss. This approach will refine sentiment labels and provide a more accurate understanding of vaccine discourse and validate LLM's labelling of comments. Furthermore, the manually attached labels can be used to fine-tune the LLM.
3. **Correlation Matrix Between Emotions, Sentiment, and Vaccination View-** In the prototype, analyses of each of the categories were done independently. However, these categories are related. For example, an anti-vaccine comment tends express fear and anger. Creating a correlation matrix to explore relationships between emotions, sentiment, and vaccination views will reveal patterns in how emotions influence pro- or anti-vaccine sentiments.
4. **Influence of YouTube's Algorithm**
   Investigating the impact of YouTube's algorithm on content recommendations will provide insights into how it amplifies vaccine discourse. Understanding the algorithm's role can help health communicators ensure that scientifically accurate content is prioritized.
5. **Vaccine hesitancy and political ideology**: Through manual readings, we observed that the politcial ideologies of the user also influenced their views about a particular brand of vaccine. Therefore, labelling political ideologies using a fine-tuned large language model could reveal any larger trends.
6. **Creating a Real-Time Sentiment Analysis Tool for YouTube**
   Developing a real-time sentiment analysis tool for YouTube comments would enable ongoing

monitoring of vaccine-related discourse. Such a tool would allow for timely responses to misinformation and shifts in public sentiment, improving public health engagement.

These directions will help refine the study and provide practical solutions for engaging with vaccine-related content on social media.

Conclusion

In conclusion, this study reveals the deeply polarized nature of vaccine discourse on YouTube, with a substantial portion of comments expressing negative emotions such as anger and fear. The findings emphasize the role of misinformation in driving vaccine hesitancy and the need for targeted public health campaigns that address both emotional and factual concerns. Moving forward, addressing the emotional landscape of vaccine discourse and combating misinformation will be crucial in improving public health outcomes in India.

Supplementary Information

The Python code, the YouTube Comments and the data visualisation have been made available on GitHub. https://github.com/ShweataNHegde/vaccine-misinfo-ngia.

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